Problem Set 3

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1 Peruvian Recycling

1.1 Recycling bin ATE

In Column 3 of Table 4A, what is the estimated ATE of providing a recycling bin on the average weight of recyclables turned in per household per week, during the six-week treatment period? Provide a 95% confidence interval. Provide a short narrative using inline R code, such as r inline_reference.

```
# Use this code chunk to show your code work (if needed)
```

Answer: ...

1.2 SMS ATE

In Column 3 of Table 4A, what is the estimated ATE of sending a text message reminder on the average weight of recyclables turned in per household per week? Provide a 95% confidence interval and provide a short narrative using inline R code.

```
# Use this code chunk to show your code work (if needed)
```

Answer: ...

1.3 What outcomes does a recycling bin affect?

Which outcome measures in Table 4A show statistically significant effects (at the 5% level) of providing a recycling bin? How are you dealing with the issue that there are several different tests that have been run, and that you are reading? How, if at all, do the authors deal with this?

Answer: ...

1.4 What outcomes does a SMS affect?

Which outcome measures in Table 4A show statistically significant effects (at the 5% level) of sending text messages? Now that you have read across two different treatments, and many outcomes, what, if anything do the p-values mean to you? Does this feel like p-hacking or doing careful investigation?

Answer: ...

1.5 Marginal effects

Suppose that, during the two weeks before treatment, household A turns in 2kg per week more recyclables than household B does, and suppose that both households are otherwise identical (including being in the same treatment group). From the model, how much more recycling do we predict household A to have than household B, per week, during the six weeks of treatment? Provide only a point estimate, as the confidence interval would be a bit complicated. This question is designed to test your understanding of slope coefficients in regression.

```
# Use this code chunk to show your code work (if needed)
```

Answer: ...

1.6 Covariates or confounders?

Suppose that the variable "percentage of visits turned in bag, baseline" had been left out of the regression reported in Column 1. What would you expect to happen to the results on providing a recycling bin? Would you expect an increase or decrease in the estimated ATE? Would you expect an increase or decrease in the standard error? Explain your reasoning.

```
# Use this code chunk to show your code work (if needed)
```

Answer: ...

1.7 Bad control or useful subset?

In column 1 of Table 4A, would you say the variable "has cell phone" is a bad control? Explain your reasoning, and engage both with the definition of a bad control, and also the implications of including a bad control in a model.

Answer: ...

1.8 What happens if you remove "has cell phone"?

If we were to remove the "has cell phone" variable from the regression, what would you expect to happen to the coefficient on "Any SMS message"? Would it go up or down? Explain your reasoning.

```
# Use this code chunk to show your code work (if needed)
```

Answer: ...

2 Multifactor Experiments

2.1 Experiment design?

What is the full experimental design for this experiment? Tell us the dimensions, such as 2x2x3. The full results appear in Panel 4B. We'll note that the dimensions of an experiment are defined in terms of the *treatments that the experiment assigns*, not in terms of other features about the data.

Answer: ...

2.2 Baseline for interpretation

In the results of Table 4B, describe the baseline category. That is, in English, how would you describe the attributes of the group of people for whom all dummy variables are equal to zero?

Answer: ...

2.3 Bin without sticker effect

In column (1) of Table 4B, interpret the magnitude of the coefficient on "bin without sticker." What does it mean?

Answer: ...

2.4 With or without a sticker?

In column (1) of Table 4B, which seems to have a stronger treatment effect, the recycling bin with message sticker, or the recycling bin without sticker? How large is the magnitude of the estimated difference?

Answer: ...

2.5 Statistical significantly different with or without a sticker?

Is this difference you just described statistically significant? Explain which piece of information in the table allows you to answer this question.

Answer: ...

2.6 Fully saturated?

Notice that Table 4C is described as results from "fully saturated" models. What does this mean? What does David Reiley propose this definition means to him in the async lecture? What do the authors seem to think it means to them? Looking at the list of variables in the table, explain in what sense the model is "saturated."

Answer: ...

3 Now! Do it with data

3.1 Treatment only model

A. For simplicity, let's start by measuring the effect of providing a recycling bin, ignoring the SMS message treatment (and ignoring whether there was a sticker on the bin or not). Run a regression of Y on only the bin treatment dummy, so you estimate a simple difference in means. Provide a 95% confidence interval for the treatment effect, using **of course** robust standard errors (use these throughout) and provide a brief narrative using R inline statements.

```
mod_basic <- 'fill this in'
## TODO, include code to print these estimates and robust standard errors to the screen
## remove these comments once you've written that code.
```

Narrative: ...

3.2 Treatment and pre-treatment values

Now add the pre-treatment value of Y as a covariate. Provide a 95% confidence interval for the treatment effect. Explain how and why this confidence interval differs from the previous one.

```
mod_pretreat <- 'fill this in'
## TODO, include code to print these estimates and robust standard errors to the screen
## remove these comments once you've written that code.
```

Answer: ...

3.3 Add state fixed effects

Now add the street fixed effects. (You'll need to use the R command `factor()` You can do this either within the `lm` call, or you can move this factoring up in the data pipeline so that it persists through the rest of your analysis. The only thing we would recommend that you *not* do is to engineer a new, persistent feature at this point.) Provide a 95% confidence interval for the treatment effect and provide a brief narrative using r inline statements.

Narrative: ...

3.4 Test for block fixed effects

Recall that the authors described their experiment as "stratified at the street level," which is a synonym for blocking by street. Does including these block fixed effects change the standard errors of the estimates *very much*? Conduct the appropriate test for the inclusion of these block fixed effects, and interpret them in the context of the other variables in the regression.

```
test_fixed_effects <- 'fill this in'
## TODO, include code to print these estimates from the appropriate test
```

Answer: ...

3.5 Feature (no) cell phone

Perhaps having a cell phone helps explain the level of recycling behavior. Instead of "has cell phone," we find it easier to interpret the coefficient if we define the variable " no cell phone." Give the R command to define this new variable, which equals one minus the "has cell phone" variable in the authors' data set. Use "no cell phone" instead of "has cell phone" in subsequent regressions with this dataset.

```
## rather than letting this feature engineering persist here -- which is
## bad practice because it requires that you keep in mind what code you
```

```
## have, and what code you have not run
##
## instead, move this recode up to the top of your data loading in this file
## so that it runs everytime that you load your data
```

Now add "no cell phone" as a covariate to the previous regression. Provide a 95% confidence interval for the treatment effect. Explain why this confidence interval does not differ much from the previous one.

```
mod_cellphone <- 'fill this in'
```

Answer: ...

3.6 Add the sms treatment

Now let's add in the SMS treatment. Re-run the previous regression with "any SMS" included. You should get the same results as in Table 4A. Provide a 95% confidence interval for the treatment effect of the recycling bin. Explain why this confidence interval does not differ much from the previous one.

```
mod_sms <- 'fill this in'
```

Answer: ...

3.7 Reproduce Table 4B, Column (2)

Now reproduce the results of column 2 in Table 4B, estimating separate treatment effects for the two types of SMS treatments and the two types of recycling-bin treatments. Provide a 95% confidence interval for the effect of the unadorned recycling bin. Explain how your answer differs from that in question 3.6, and explain why you think it differs.

Answer: ...

4 A Final Practice Problem

4.1 Simple treatment effect of Zmapp

Without using any covariates, answer this question with regression: What is the estimated effect of ZMapp (with standard error in parentheses) on whether someone was dehydrated on day 14? What is the p-value associated with this estimate?

```
zmapp_1 <- 'fill this in'
```

Answer: ...

4.2 Add baseline covariates

Add covariates for dehydration on day 0 and patient temperature on day 0 to the regression from part (a) and report the ATE (with standard error). Also report the p-value.

```
zmapp_2 <- 'fill this in'
```

Answer: ...

4.3 Interpret estimates

Do you prefer the estimate of the ATE reported in the chunk called `dehydration model` or `add pre-treatment measures`? Why? Report the results of the F-test that you used to form this opinion.

```
zmapp_test_object <- 'fill this in'
```

Answer: ...

4.4 Add day fourteen temperature

The regression from part `add pre-treatment measures` suggests that temperature is highly predictive of dehydration. Add, temperature on day 14 as a covariate and report the ATE, the standard error, and the p-value.

```
zmapp_3 <- 'fill this in'
```

Answer: ...

4.5 Interpret estimates

Do you prefer the estimate of the ATE reported in part `add pre-treatment measures` or `add day 14 temperature`? What is this preference based on?

Answer: ...

4.6 Look at temperature

Now let's switch from the outcome of dehydration to the outcome of temperature, and use the same regression covariates as in the chunk titled `add pre-treatment measures`. Test the hypothesis that ZMapp is especially likely to reduce mens' temperatures, as compared to womens', and describe how you did so. What do the results suggest?

```
zmapp_4 <- 'fill this in'
```

Answer: ...

4.7 Compare health outcomes

Which group – those that are coded as `male == 0` or `male == 1` have better health outcomes (temperature) in control? What about in treatment? How does this help to contextualize whatever heterogeneous treatment effect you might have estimated?

Answer: ...

4.8 Collaborating with others, Part (1)

Suppose you speak with a colleague to learn about heterogeneous treatment effects.

This colleague has access to a non-anonymized version of the same dataset and reports that they looked at heterogeneous effects of the ZMapp treatment by each of 80 different covariates to examine whether each predicted the effectiveness of ZMapp on each of 20 different indicators of health.

Across these regressions your colleague ran, the treatment's interaction with sex on the outcome of temperature is the only heterogeneous treatment effect that he found to be statistically significant. They reason that this shows the importance of sex for understanding the effectiveness of the drug, because nothing else seemed to indicate why it worked. Bolstering your colleague's confidence, after looking at the data, they also returned to their medical textbooks see the whispers of a theory about why ZMapp interacts with processes only present in men to cure.

Another doctor, unfamiliar with the data, hears your colleague's theory and finds it plausible. How likely do you think it is ZMapp works especially well for curing Ebola in men, and why? (This question is conceptual can be answered without performing any computation.)

Answer: ...

4.9 Collaborating with others, Part (2)

Suppose that your colleague conducted their research looking at the interaction of 80 covariates with ZMapp, but that you on your own tested this and only this HTE, and discovered a positive result. How, if at all, does your colleague's behavior change the interpretation of your test? Does this seem fair or reasonable?

Answer: ...

4.10 Collaborating with others, Part (3)

Now, imagine that your colleague had not conducted the 80 different regressions. Instead, they tested this heterogeneous treatment effect, and only this heterogeneous treatment effect, of their own accord. Would you be more or less inclined to believe that the heterogeneous treatment effect really exists? Why? Is there a general principle that is guiding your reasoning?

Answer: ...